

Economics G6410 - Mathematical Methods for Economists - Fall 2019
Math Camp Exam

Department of Economics

Columbia University

Date: 6:10 - 8:00 p.m., September 3rd, 2019

Instructions:

This is a closed-book, closed-note exam, and no calculator is allowed. There are 10 questions in total, and the full grade is 100 points.

1. **(10 pts)** Let (X, d) be a metric space. Let $S \subset X$ be a finite set. Prove that S is closed, bounded, and compact.

• **Solution:** Let $S = \{x_1, \dots, x_n\}$ be a finite set. (i) S is closed because any finite set does not have a limit point, so that S is closed is vacuously true. (ii) Pick any arbitrary point $x \in X$. Let $r = \max\{d(x_1, x), d(x_2, x), \dots, d(x_n, x)\} + 1$. Since $S \subset B_r(x)$, S is bounded. (iii) Finally, S is compact since for any arbitrary open cover of S , we can find a finite subcover of S . We can always do this by considering one open set covering each x_i , $i = 1, \dots, n$, and taking union of them. $\square$

2. **(10 pts)** Let $f : C \rightarrow \mathbb{R}$ be a continuous *bounded* function whose domain C is a *closed* subset of $\mathbb{R}^n$. Does f necessarily achieve the supremum of its values?

• **Solution:** No. Here is a counterexample. Let $C = [1, \infty) \subset \mathbb{R}$. Note that C is closed because it has all of its limit points. Let $f : C \rightarrow \mathbb{R}$ be defined as $f(x) = -1/x$. Clearly, f is bounded in $[-1, 0]$. However, f does not achieve its supremum because $f(x)$ is increasing, and $\lim_{x \rightarrow \infty} f(x) = 0$ so 0 is the supremum, but $f(x) < 0$ for any $x \in C$. $\square$

3. **(10 pts)** Consider the following matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

Is A diagonalizable in $\mathbb{C}$? Is it diagonalizable in $\mathbb{R}$? If so, diagonalize it, i.e., find a 2×2 invertible matrix P s.t. $P^{-1}AP = \Lambda$, where Λ is a diagonal matrix.

• **Solution:** Consider the characteristic polynomial of A .

$$\det(A - \lambda I_2) = \det \begin{bmatrix} -\lambda & 1 \\ 1 & 1 - \lambda \end{bmatrix} = (-\lambda)(1 - \lambda) - (1)(1) = \lambda^2 - \lambda - 1 = 0$$

The roots are

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$

Set $Ax = \lambda_i x$ for $i = 1, 2$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda_i \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \implies \begin{bmatrix} x_2 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} \lambda_i x_1 \\ \lambda_i x_2 \end{bmatrix}$$

We can find that the eigenvectors associated with λ_1 and λ_2 are $v_1 = (2, 1 + \sqrt{5})$ and $v_2 = (2, 1 - \sqrt{5})$ respectively. By definition, A is diagonalizable in $\mathbb{R}$. Moreover, $P = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$ where v_1 and v_2 are two eigenvectors represented as column vectors. $\square$

4. (10 pts) Approximate $f(x) = e^{x+1}$ at $x_0 = 0$ with a Taylor polynomial of order three.

• **Solution:** Recall that $f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{6}(x - x_0)^3$. We have $f(x) = f'(x) = f''(x) = f'''(x) = e^{x+1}$, so $f(x_0) = f'(x_0) = f''(x_0) = f'''(x_0) = e^{x_0+1}$. Then

$$f(x) \approx e + ex + \frac{e}{2}x^2 + \frac{e}{6}x^3$$

$\square$

5. (10 pts) Prove that any collection of vectors that includes the zero-vector cannot be linearly independent.

• **Solution:** Zero vector is always linearly dependent, because $\lambda \mathbf{0} = \mathbf{0}$ does not imply $\lambda = 0$. Then for any collection of vectors that includes the zero-vector, say $S = \{\mathbf{0}, v_1, v_2, \dots, v_n\}$,

$$\lambda_0 \mathbf{0} + \lambda_1 v_1 + \dots + \lambda_n v_n = \mathbf{0}$$

does not imply $\lambda_0 = \lambda_1 = \dots = \lambda_n = 0$, so they are not linearly independent. $\square$

6. (10 pts) Prove the following statements:

- (a) In $(\mathbb{R}, d_2)$, let $\{x_n\}_{n=1}^\infty$ be a convergent sequence with limit x , and let b be a number such that $x_n \leq b$ for all n . Then $x \leq b$.
- (b) If $x \in \text{int}S$ and $\{x_n\}_{n=1}^\infty$ is a sequence converging to x , then $x_n \in S$ for all sufficiently large n .

• **Solution:**

- (a) Suppose that $x_n \leq b$ for all n , but that $x > b$. Choose ε so that $0 < \varepsilon < x - b$; then, $b < x - \varepsilon$ and $I_\varepsilon(x) = (x - \varepsilon, x + \varepsilon)$ lies to the right of b on the number line. There is an integer N such that for all $n \geq N$, $x_n \in I_\varepsilon(x)$. For these x_n 's, $b < x_n$; this is a contradiction to the hypothesis that all the x_n 's were $\leq b$. $\square$
- (b) That $x \in \text{int}S$ means that we can find some $\varepsilon > 0$ such that $B_\varepsilon(x) \subset S$. Since $x_n \rightarrow x$, for the chosen $\varepsilon > 0$, we can find $N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for all $n > N$. Combining these directly implies the conclusion we want. $\square$

7. (10 pts) For each of the following subsets of $(\mathbb{R}^2, d_2)$ (plane), draw the set, state whether it is open, closed, neither, or both, and justify your answer in a word or two:

- (a) $\{(x, y) : -1 < x < 1, y = 0\}$
- (b) $\{(x, y) : x \text{ and } y \text{ are integers}\}$
- (c) $\{(x, y) : x + y = 1\}$
- (d) $\{(x, y) : x + y < 1\}$
- (e) $\{(x, y) : x = 0 \text{ or } y = 0\}$

(f) $\{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$

• **Solution:**

- (a) Not open because $(x, y) = (0, 0)$ is not an interior point of the set. Not closed because $(x, y) = (1, 0)$ is a limit point of the set which is not included in the set.
- (b) Not open because any point in the set is not an interior point of the set. Closed because there is no limit point of the set.
- (c) Not open because any point in the set is not an interior point of the set. Closed because it contains all of its limit points (which are the whole set).
- (d) Open because all points in the set are interior points of the set. Not closed because $(x, y) = (1, 0)$ is a limit point not included in the set.
- (e) Not open because no point in the set is an interior point of the set. Closed because it contains all of its limit points.
- (f) Open and closed because the set is the entire space.

□

8. (10 pts) Let $(\mathbb{R}^n, d_2)$ be a metric space. Prove that the intersection of compact sets is compact and that the finite union of compact sets is compact. Show that the infinite union of compact sets need not be compact.

• **Solution:**

- Let $\{S_\alpha\}_{\alpha \in A}$ be an arbitrary family of compact sets in $(\mathbb{R}^n, d_2)$. Consider $S \equiv \cap_{\alpha \in A} S_\alpha$. We want to show that S is compact. Since (i) S_α is closed and bounded for each α (Heine-Borel Theorem), (ii) arbitrary intersection of closed sets is closed, (iii) arbitrary intersection of bounded sets is bounded, it follows that $\cap_{\alpha \in A} S_\alpha$ is closed and bounded, which implies compactness of S .
- Let $\{S_\alpha\}_{\alpha \in A}$ be a finite collection of compact sets in $(\mathbb{R}^n, d_2)$. Consider $S = \cup_{\alpha \in A} S_\alpha$. Since finite union of closed sets is closed, and finite union of bounded sets is bounded, it follows that $\cup_{\alpha \in A} S_\alpha$ is closed and bounded.
- Let us provide a counterexample. Let $S_k = [k, k + 1]$ for each $k \in \mathbb{Z}$ which is clearly closed and bounded, and consider $S = \cup_{k \in \mathbb{Z}} S_k$. Clearly, $S = (-\infty, \infty)$ which is not bounded, hence not compact.

□

9. (10 pts) Determine whether the following statement is true or false. If it is true, prove it. If not, provide a counterexample: Let (X, d_X) and (Y, d_Y) be metric spaces, and function $f : X \rightarrow Y$. The function f is a continuous function iff $f(E)$ is open in (Y, d_Y) for any set E open in (X, d_X) .

- **Solution:** The statement is false. Let $S = (-\infty, \infty)$ which is open in $(\mathbb{R}, d_2)$, and $f(x) = x^2$ which is continuous. Then $f(S) = [0, \infty)$ which is not open. □

10. (10 pts) Consider the following problem:

$$\max_{(x,y) \in \mathbb{R}^2} x^2 + 4y^2$$

subject to

$$2x + 2y \leq 1 \tag{1}$$

$$x \geq 0 \tag{2}$$

$$y \geq 0 \tag{3}$$

- (a) Does a solution exist?
- (b) Should constraint (1) be binding at any solution?
- (c) Solve the program.

• **Solution:**

- (a) The solution exists by the Weierstrass Theorem because $x^2 + 4y^2$ is continuous and the constraints define a compact set. Since we are looking for optima of continuous functions at compact domain, the solution must exist.
- (b) Yes. The objective function $x^2 + 4y^2$ is monotonically increasing in x and y . So if $2x + 2y \leq 1$ is not binding, we can always attain a higher value for the objective function. Formally, if $2x^* + 2y^* < 1$ at the optimum, we can always do better by considering $x' = x^* + \varepsilon$ for small enough ε such that $2(x' + \varepsilon) + 2y^* < 1$, but this gives us $(x' + \varepsilon)^2 + 4y^{*2} > x^{*2} + 4y^{*2}$.
- (c) Note that we do not even have to use KKT conditions. By the argument in (b), $2x + 2y = 1$ must hold at the optimum, but this in turn shows the trade-off relationship between x and y . It should be clear from the objective function that it is optimal to have $x = 0$ and $y = \frac{1}{2}$. Hence the solution is $(x^*, y^*) = (0, \frac{1}{2})$. Other characterizations, e.g., KKT conditions, can be used to obtain the same result.